Classifying Heart Disease from BRFSS Data

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***Abstract*—We compare least squares, ridge, and logistic regression for classifying a reported history of myocardial infarction or coronary heart disease from the 2015 BRFSS survey. The training data has 328,135 respondents, of whom 8.83% belong to the positive class. After handling coded and missing answers, we evaluate least squares, ridge, logistic regression, and regularized logistic regression using stratified five-fold cross-validation, preserving original class proportions within each fold. Throughout our model evaluations, we use decision thresholds selected on the inner training splits. Our selected unregularized logistic regression model achieves a mean validation F_1 of 0.4275 (standard deviation 0.0031) and a test (AIcrowd) F_1 score of 0.439. The biggest performance gains came from correctly interpreting survey answers and choosing the best decision thresholds. On the other hand, regularization and class rebalancing provided no clear F1 improvement.**

***Index Terms*—Heart disease, BRFSS, Logistic regression, Cross-validation, Class imbalance**

I. Introduction

In 2015, the Behavioral Risk Factor Surveillance System (BRFSS) collected responses about demographics, lifestyle, and health. In our study, we investigate whether these survey responses can help identify people who report a history of myocardial infarction or coronary heart disease. To do this, we compare different regression-based classification methods, examining how preprocessing and model evaluation choices affect their performance.

Initial data exploration (see `Data_Exploration.ipynb`) revealed that accuracy alone is a poor guide to performance because of class imbalance. Predicting every respondent as negative would achieve 91.17% accuracy without detecting a single positive case. We therefore focus on $F_1 = \frac{2PR}{P+R}$, where P denotes precision and R denotes recall.

Some features also contain missing answers and numeric codes that require different treatment. For example, "none" represents a valid zero, while "don't know" is treated as missing information. Treating both as ordinary numbers would be a mistake, changing the meaning of the responses.

II. Methods

The survey is not a clean numeric table, and the official codebook [1] is essential to read it. Our pipeline (in `preprocessing.py`) applies five steps.

Excluding columns (103 of 321). We drop 31 administrative variables (identifiers, dates, interview mode, survey weights), 2 date columns, and the state code. We also drop **one leakage column**: `HAREHAB1` (cardiac rehabilitation) is only asked to people who reported a heart attack, which is part of the definition of the target. Finally, we drop 68 largely **redundant** columns that are derived from or overlap substantially with other columns.

Recoding special codes. In BRFSS, "don't know" and "refused" are stored as 7, 9, 77, 99, 777 or 999, and "none"/"never" as 88, 555 or 888. We map the first group to missing and the second to 0, with the rule chosen by the range of each column and a list of exceptions checked one by one in the codebook (drink counts 1–76 are valid, while 77 means "don't know"; 88 is a valid age at diabetes diagnosis).

Handling missing values. Every respondent has at least 25% missing answers (median 43%), so rows cannot be dropped; we work column-wise. Most missingness is not random: it comes from **skip patterns**, questions asked only to a subgroup (blood-pressure medication only to hypertensive people). We keep the columns with at most a fraction τ of missing values, impute the median, and add one binary indicator per distinct missingness pattern. We selected τ over the grid $0.1, 0.3, 0.5, 0.8, 0.9$ by cross-validation (F1 from 0.4133 to 0.4258, AUC from 0.8517 to 0.8589, improving overall); $\tau = 0.9$ was best but within 0.6 SE of $\tau = 0.8$ on F1, so by the rule below we kept the smaller, simpler model.

Encoding and scaling. For 23 variables the numeric code carries no order (marital status) or an order that does not match the risk (`_SMOKER3`: former smokers are at higher risk than current ones), so we one-hot encode them, using $k-1$ dummies to avoid collinearity with the bias. All features are then standardised with the training statistics.

With $\tau = 0.8$ this turns 106 survey questions into **277 features** (83 numeric, 95 dummies, 98 missingness indicators and the bias).

The decision threshold is itself a hyperparameter, to select it, we used **stratified 5-fold cross-validation** (seed 42). In each fold, we pick the F_1-maximising threshold on an inner fifth of the training part, refit on the whole training part, and evaluate on the held-out fold. Each threshold is chosen without using the held-out fold, whose results are then used to compare models.

To compare two configurations, we use their **paired** per-fold differences, which cancel the variability due to some folds being easier than others. The five folds are not independent (their training sets overlap by 75%), so we correct the standard error [2]:

$$\widehat{\mathrm{SE}} = \sqrt{\left(\frac{1}{k} + \frac{n_{\text{test}}}{n_{\text{train}}}\right) s_d^2}, \quad \frac{n_{\text{test}}}{n_{\text{train}}} = \frac{1}{k-1}. \tag{1}$$

We also report AUC, the probability that a random patient is ranked above a random non-patient. AUC does not depend on the threshold and is much less noisy than F1, which makes it the better detector of small real improvements. **All decision rules were fixed before looking at the numbers:**

a modification is adopted only if it improves F1 by at least 2 SE **and** 0.002, without lowering AUC by more than 2 SE.

Least squares, ridge, logistic and regularized logistic regression all use the same five-fold protocol. To verify convergence, GD and SGD are checked against the least-squares solution on fold 0, using 3,000 and 3,000,000 updates respectively. The final logistic model starts from $w = 0$ and uses 1000 full-batch updates with $\gamma = 1/L$, where $L = \lambda_{\max}(X^\top X/n)/4$. On fold 0, increasing this to 2000 updates changes validation AUC by $3 \cdot 10^{-4}$.

III. Results

TABLE I
Cross-validated performance.

Model	F1	AUC
Always positive	0.162	0.500
Logistic (age, sex only)	0.294	0.753
Least squares (GD/SGD converge to it)	0.4140	0.8518
Ridge, $\lambda = 0.1$	0.4142	0.8529
Logistic	**0.4275**	**0.8600**
Regularized logistic, $\lambda = 10^{-4}$	0.4271	0.8600

From the results, we see that logistic regression performs better than least squares, with an improvement of 0.0135 in F_1 and 0.0082 in AUC (Table I). We tested λ values from 10^{-6} to 10^{1} for ridge and from 10^{-6} to 10^{-2} for regularized logistic regression, in powers of ten. Ridge regression increased least-squares AUC by only 0.0011, while regularized logistic regression performed almost identically to unregularized logistic regression. Since regularization provided no clear improvement in F_1, we retained the simpler unregularized logistic regression model.

Neither weighting positive samples by $\alpha \in \{2, 5, 10.3\}$ nor retaining negatives with probability $1/\alpha$ improved mean validation F_1. Weighting increased the selected thresholds from 0.196 to 0.333, 0.554, and 0.734, while AUC changed only slightly. The strongest negative undersampling reduced AUC.

On fold 0, training on subsets from 5% to 100% reduced the train–validation AUC gap, reaching 0.8611 and 0.8612 respectively at full training size. Preprocessing was fitted once on the full training fold and held fixed across subsets. We then tested non-linear features. Quintile-binning continuous variables barely changed AUC (+0.0001). Interacting all features with age (552 features, excluding the bias) improved AUC by 0.0014 (4.9 SE), but the F_1 gain (+0.0010, 0.6 SE) was too small to meet our selection criterion. Sex interactions provided no further F_1 improvement.

TABLE II
Ablation: one component removed at a time from the final model.

Component removed	ΔF1	ΔAUC
Tuned threshold (0.5 instead)	−0.1939	0
Recoding of special codes	−0.0173	−0.0111
Logistic loss (least squares)	−0.0135	−0.0082
One-hot encoding	−0.0025	−0.0013
$\tau = 0.8$ ($\tau = 0.3$ instead)	−0.0020	−0.0007
Missingness indicators	−0.0000	−0.0005

Removing one step at a time (Table II) shows that tuning the threshold is critical: using $t = 0.5$ gives accuracy 0.9148 and $F_1 = 0.2336$, compared with accuracy 0.8701 and $F_1 = 0.4275$ using tuned thresholds. Recoding special survey codes also matters: omitting it reduces mean validation F_1 by 0.0173.

IV. Conclusion

In conclusion, we retain unregularized logistic regression with a submission threshold of $t = 0.20$, achieving a test (AIcrowd) F_1 score of 0.439. The largest changes in performance came from threshold selection and the transformation of survey codes into usable numbers. Having the threshold at 0.5 reduces mean validation F1 from 0.4275 to 0.2336, while omitting the recoding step reduces it to 0.4102. We saw that regularization, class weighting, and undersampling provide no clear F1 gains. Finally, interactions with age improve AUC, but their F1 improvement does not meet our selection criterion. We therefore keep the simpler model.

Reproducibility. `python run.py` with the dataset in `dataset/` rebuilds the submission (seed 42, no randomness other than the fold split). The notebook `project1_notebook.ipynb` contains every experiment reported here; all of them import the same `preprocessing.py` and `validation.py`.

References

[1] Centers for Disease Control and Prevention, "BRFSS 2015 Codebook Report," technical report, 2016.

[2] C. Nadeau and Y. Bengio, "Inference for the Generalization Error," *Machine Learning*, vol. 52, no. 3, 2003.